

STABILITY OF 3-PACKED WORDS IN THE PLACTIC MONOID

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ABSTRACT. Let $C(u)$ be the centralizer of a word u modulo Knuth equivalence. We prove that every word over $\{1, 2, 3\}$ in which all three letters occur is 3-stable: $C(u^k) = C(u^3)$ for every $k \geq 3$, where the words in the centralizers may use arbitrary positive-integer letters. The internal rank-three part follows from the faithful tropical representation of the rank-three plactic monoid. We prove that, inside the interval-subsequence cone, commutation with A^k is independent of $k \geq 2$; the proof reduces the corner entry to the thirteen weak orders of the diagonal. A separate column-factorization lemma isolates letters larger than three. We also give an independently verified exact computation reproducing the previously reported finite range and testing three larger, separately stated ranges.

1. INTRODUCTION

Write $x \equiv y$ when the words x, y are Knuth equivalent, or equivalently when their row-insertion tableaux are equal. The plactic centralizer of u is

$$C(u) = \{w : uw \equiv wu\}.$$

A word is K -stable if $C(u^k) = C(u^K)$ for all $k \geq K$. Sagan and Zhao proved strong stability for binary words and m -stability for permutations in $\mathfrak{S}_m$, and conjectured that every m -packed word is m -stable [2]. In the locally archived arXiv source this is Conjecture 3.8; Conjecture 4.6 there is instead a log-concavity conjecture.

The first case not covered by those theorems is the alphabet [3]. We settle it here. The quantification over w in the following theorem is over all finite words on the positive integers, not merely words over [3].

Theorem 1.1. *Let $u \in \{1, 2, 3\}^*$ and suppose that each of 1, 2, 3 occurs in u . Then*

$$C(u^k) = C(u^3) \quad (k \geq 3).$$

Thus every 3-packed word is 3-stable.

There are two distinct reasons for the two thresholds appearing in the proof. Within the rank-three plactic monoid, a tropical matrix argument shows that the relevant commutation conditions are already independent of $k \geq 2$. For a centralizer witness using larger letters, three copies of a 3-packed

word provide the decreasing subsequence 321; a published row bound then separates the first three tableau rows from the remaining tail.

2. TABLEAUX AND SEPARATION OF THE LARGE-LETTER TAIL

Let $\mathbb{P} = \{1, 2, \dots\}$ and include the empty word in $\mathbb{P}^*$. We use semistandard tableaux with weakly increasing rows and strictly increasing columns. For a word w , let $P(w)$ be its row-insertion tableau and let $w_{\leq m}$ denote the subword obtained by erasing all letters larger than m . The standard restriction property is

$$(1) \quad P(w_{\leq m}) = P(w)_{\leq m}.$$

We use three results recorded by Sagan and Zhao [2]. First, if a has maximum letter m , contains the decreasing subsequence $m(m-1)\cdots(m-r+1)$, and $w \in C(a)$, then every entry in each of the first r rows of $P(w)$ is at most m (their Proposition 2.4). Second, if $\delta_m = m(m-1)\cdots 1$ and $v \in [m]^*$, then left multiplication by δ_m prepends an i to row i of $P(v)$ (their Lemma 3.3). In particular, left multiplication by δ_m^t is injective on Plac_m . Third, a word w commutes with δ_m if and only if the first m rows of $P(w)$ contain no letter larger than m (their Proposition 3.6).

The next lemma is the bridge from arbitrary witnesses to rank three.

Lemma 2.1 (tail isolation). *Let $v \in [3]^*$ and $w \in \mathbb{P}^*$. Suppose every entry in the first three rows of $P(w)$ is at most 3. Then*

$$vw \equiv wv \iff vw_{\leq 3} \equiv w_{\leq 3}v.$$

Proof. Use the column-reading word of $P(w)$, reading each column from bottom to top and the columns from left to right. The columns of height at least four are the leftmost columns. Thus

$$(2) \quad w \equiv \mathcal{C}D, \quad \mathcal{C} = C_1 \cdots C_t,$$

where the C_j are the readings of the tall columns and D is the reading of the remaining columns. The row hypothesis and column strictness force the top three entries of every tall column to be 1, 2, 3. Hence $C_j = z_j 321$, where all letters of the decreasing word z_j are larger than 3. The word D lies in $[3]^*$.

We use a direct complete-column calculation. If the entries of a tall column are $c_1 < c_2 < \cdots < c_h$ and $\Delta = c_h c_{h-1} \cdots c_1$ is its reading, then literal row insertion gives, for every j ,

$$P(\Delta c_j) = P(c_j \Delta) = \begin{array}{cc} c_1 & c_j \\ c_2 & \\ \vdots & \\ c_h & \end{array}.$$

For the first product, c_j appends to the first row of the column tableau. For the second, first suppose $j < h$. After inserting $c_j, c_h, \dots, c_{j+1}$ the nonempty rows are $[c_j, c_{j+1}], [c_{j+2}], \dots, [c_h]$. The next letter, which is the copy of c_j inside Δ , bumps the right entry of the first row and cascades down this single

chain. Thereafter, immediately after inserting c_r for $r = j - 1, j - 2, \dots, 1$, the rows are $[c_r, c_j], [c_{r+1}], \dots, [c_h]$.

If $j = h$, the first two letters are equal and append to form $[c_h, c_h]$. For $r = h - 1, h - 2, \dots, 1$, insertion of c_r then changes the rows to $[c_r, c_h], [c_{r+1}], \dots, [c_h]$. This also covers $h = j = 1$. Thus the displayed tableau holds for every j , including both endpoints. Hence Δ commutes with every generator in its support, and therefore with every word on that support. Since the support of each C_j contains 1, 2, 3, we obtain

$$(3) \quad vC_j \equiv C_jv, \quad v\mathcal{C} \equiv \mathcal{C}v.$$

Left multiplication by $\mathcal{C}$ is injective on rank-three plactic classes. Indeed, the first three rows of $P(\mathcal{C})$ form a $3 \times t$ rectangle with entries i in row i . When a word $X \in [3]^*$ is inserted on the right, no bumping path passes below row three: an entry reaching row three is at most 3 and appends after the existing 3's. The lower rows are therefore unchanged, while the first three rows are $P((321)^t X)$. Lemma 3.3 cited above says that these rows are obtained from $P(X)$ by prepending t copies of i to row i . Deleting those copies recovers $P(X)$. Consequently

$$(4) \quad \mathcal{C}X \equiv \mathcal{C}Y \implies X \equiv Y \quad (X, Y \in [3]^*).$$

Equations (2)–(4) give

$$vw \equiv wv \iff \mathcal{C}vD \equiv \mathcal{C}Dv \iff vD \equiv Dv.$$

On the other hand, erasing every letter greater than 3 from (2), and using (1), gives $w_{\leq 3} \equiv (321)^t D$. The element $(321)^t$ is central and cancellable in Plac_3 , so

$$vw_{\leq 3} \equiv w_{\leq 3}v \iff vD \equiv Dv.$$

The two equivalences prove the lemma, including $t = 0$. $\square$

3. THE FAITHFUL TROPICAL REPRESENTATION

Work over the max-plus tropical semiring $\mathbb{T} = \mathbb{R} \cup \{-\infty\}$. For a nonempty word $a \in [3]^*$, define the upper-triangular matrix $\phi_3(a)$ by

$$\phi_3(a)_{ij} = \max \left\{ |s| : \begin{array}{l} s \text{ is a weakly increasing subsequence of } a \\ s \text{ uses only letters in } [i, j] \end{array} \right\} \quad (i \leq j).$$

Cain, Klein, Kubat, Malheiro, and Okniński proved that this is a plactic homomorphism [1]. They define an antihomomorphism f_3 by

$$f_3(1) = 32, \quad f_3(2) = 31, \quad f_3(3) = 21$$

and reversal of the order of factors, and an antiautomorphism π_3 of upper-triangular tropical matrices by reflection across the antidiagonal. Then

$$\sigma_3 = \pi_3 \circ \phi_3 \circ f_3, \quad \Psi_3 = (\phi_3, \sigma_3) : \text{Plac}_3 \longrightarrow U_3(\mathbb{T}) \times U_3(\mathbb{T})$$

is a faithful homomorphism [1, Proposition 5.3].

We record the additional inequalities satisfied by the matrices in its image. Write

$$(5) \quad A = \begin{pmatrix} x & d & f \\ -\infty & y & e \\ -\infty & -\infty & z \end{pmatrix}.$$

Every interval-subsequence matrix $A = \phi_3(a)$ satisfies

$$(6) \quad \begin{aligned} x, y, z &\geq 0, \\ \max(x, y) &\leq d \leq x + y, & \max(y, z) &\leq e \leq y + z, \\ \max(d, e) &\leq f \leq x + y + z, & d + e &\geq y + f. \end{aligned}$$

Only the last inequality needs comment. Choose positions $0 \leq r \leq s \leq |a|$ that realize f by taking all 1s in the prefix through r , all 2s between the two cuts, and all 3s after s ; any maximizing $1^*2^*3^*$ subsequence can be enlarged to this form. A candidate counted by d takes those prefix 1s and all 2s after the first cut. A candidate counted by e takes all 2s before the second cut and those suffix 3s. Their total counts the chosen length f once and every one of the y occurrences of 2 once more. Hence $d + e \geq y + f$. The cone (6) is invariant under antidiagonal reflection, so it contains both components of $\Psi_3(a)$.

4. CENTRALIZERS OF POWERS IN THE INTERVAL-SUBSEQUENCE CONE

The following matrix theorem is the main calculation.

Theorem 4.1. *Let A have the form (5) and satisfy (6). Let B be another upper-triangular matrix of the same form whose finite entries also satisfy (6). For any integers $k, \ell \geq 2$,*

$$A^k B = B A^k \iff A^\ell B = B A^\ell.$$

Proof. Put

$$p = \max(x, y), \quad q = \max(y, z), \quad M = \max(x, y, z), \quad h = \max(x, z),$$

and

$$(7) \quad F_2 = \max(f + h, d + e), \quad u = d - p, \quad v = e - q, \quad s = F_2 - 2M.$$

A tropical path from state 1 to state 2 uses d once and puts every other step on the better of the x, y loops; the 2, 3 entry is symmetric. A path from state 1 to state 3 either uses f , or uses d and e . Thus, for $k \geq 2$,

$$(8) \quad A^k = \begin{pmatrix} kx & kp + u & kM + s \\ -\infty & ky & kq + v \\ -\infty & -\infty & kz \end{pmatrix}.$$

For the corner, before simplification the maximum is

$$\max(f + (k - 1)h, d + e + (k - 2)M).$$

If $M = h$, this is $F_2 + (k - 2)M$. If $M = y > h$, the last inequality in (6) gives $d + e \geq f + y$, so the second term wins. This proves (8) in all cases.

Write

$$B = \begin{pmatrix} X & D & G \\ -\infty & Y & E \\ -\infty & -\infty & Z \end{pmatrix}.$$

The $(1, 2)$ entries of $A^k B$ and BA^k are equal precisely when

$$(9) \quad \max(kx + D, kp + u + Y) = \max(X + kp + u, D + ky).$$

After factoring the larger diagonal slope, this is equivalent to the following condition, independent of k :

$$(10) \quad \begin{array}{l|l} x > y & \max(D, u + Y) = X + u \\ x = y & \max(D, u + Y) = \max(X + u, D) \\ x < y & Y + u = \max(X + u, D). \end{array}$$

For example, when $x > y$, subtracting kx turns the right side of (9) into $\max(X + u, D - k(x - y))$; the decreasing term is strictly less than D , which occurs on the left. The remaining cases are symmetric or tied. Likewise, equality in position $(2, 3)$ is equivalent to

$$(11) \quad \begin{array}{l|l} y > z & \max(E, v + Z) = Y + v \\ y = z & \max(E, v + Z) = \max(Y + v, E) \\ y < z & Z + v = \max(Y + v, E). \end{array}$$

It remains to show that the corner condition is also independent of k . After subtracting kM , it is $L_k = R_k$, where

$$(12) \quad \begin{aligned} L_k &= \max(G - k(M - x), u + E - k(M - p), s + Z), \\ R_k &= \max(X + s, D + v - k(M - q), G - k(M - z)). \end{aligned}$$

The thirteen weak orders of x, y, z reduce (12) as follows. Every displayed equality is independent of k ; “automatic” means that (10)–(11) force both sides to have the stated value.

weak order	equivalent corner condition
$x > y > z$	$\max(G, u + E, s + Z) = X + s$
$x > z > y$	$\max(G, u + E, s + Z) = X + s$
$y > x > z$	automatic, value $Y + s$
$y > z > x$	automatic, value $Y + s$
$z > x > y$	$s + Z = \max(X + s, D + v, G)$
$z > y > x$	$s + Z = \max(X + s, D + v, G)$
$x = y = z$	$\max(G, u + E, s + Z) = \max(X + s, D + v, G)$
$x = y > z$	$\max(G, u + E, s + Z) = \max(X + s, D + v)$
$x = y < z$	$s + Z = \max(X + s, D + v, G)$
$x = z > y$	$\max(G, u + E, s + Z) = \max(X + s, D + v, G)$
$x = z < y$	automatic, value $Y + s$
$y = z > x$	$\max(u + E, s + Z) = \max(X + s, D + v, G)$
$y = z < x$	$\max(G, u + E, s + Z) = X + s.$

For completeness, we give all inequalities used to delete the transient terms. From (7),

$$(13) \quad s \geq p + q + u + v - 2M.$$

If $y = M$, submodularity yields $F_2 = d + e$; hence

$$(14) \quad p = q = M, \quad s = u + v.$$

We now audit the three possible maximal-diagonal families explicitly.

If $M = x$, then $p = M$, so all three terms of L_k are constant. Whenever $q < M$, condition (10) gives $D \leq X + u$ and, with $\Delta = M - q > 0$, (13) gives

$$(15) \quad D + v - k\Delta \leq X + u + v - k\Delta \leq X + s \quad (k \geq 1).$$

Whenever $z < M$, the remaining endpoint satisfies

$$(16) \quad G - k(M - z) < G \leq L_k.$$

Thus the two strict x -maximal orders and the walls $x = y > z$ and $y = z < x$ have exactly the surviving terms displayed in the table; on the wall $x = z > y$ every slope in (12) is zero.

If $M = z$, the three terms of R_k are constant. Whenever $p < M$, condition (11) gives $E \leq Z + v$ and, with $\Delta = M - p > 0$, (13) gives

$$(17) \quad u + E - k\Delta \leq u + Z + v - k\Delta \leq s + Z.$$

Whenever $x < M$, the other endpoint satisfies

$$(18) \quad G - k(M - x) < G \leq R_k.$$

This proves the two strict z -maximal rows and the walls $x = y < z$ and $y = z > x$; again $x = z > y$ has only constant terms.

Finally suppose y is the unique maximum, including the wall $x = z < y$. Equation (14) holds. The two adjacent conditions give the exact equalities

$$(19) \quad \begin{aligned} \max(X + s, D + v) &= v + \max(X + u, D) = Y + u + v = Y + s, \\ \max(u + E, s + Z) &= u + \max(E, v + Z) = Y + u + v = Y + s. \end{aligned}$$

They also give $D \leq Y + u$ and $E \leq Y + v$. Since the cone inequality for B is $D + E \geq Y + G$,

$$(20) \quad G \leq D + E - Y \leq Y + u + v = Y + s.$$

Both endpoint transients are strictly below G , so $L_k = R_k = Y + s$ in all three unique-middle-maximum orders. If $x = y = z$, every slope is zero. The preceding paragraphs cover all thirteen weak orders and import every transient-term and forced-common-value inequality; no unlisted subcase remains.

The diagonal entries of the two products agree automatically, the adjacent conditions (10)–(11) are independent of k , and the table proves the same for the corner. Hence commutation for one $k \geq 2$ is equivalent to commutation for every $k \geq 2$. $\square$

Corollary 4.2. *Let a be a nonempty word over $[3]$ and let $b \in [3]^*$. For any $k, \ell \geq 2$,*

$$a^k b \equiv b a^k \iff a^\ell b \equiv b a^\ell.$$

Proof. The empty word b is immediate. Otherwise apply Theorem 4.1 to $A = \phi_3(a), B = \phi_3(b)$ and to $A = \sigma_3(a), B = \sigma_3(b)$. All four matrices lie in (6). The two resulting matrix equalities are equivalent to the plactic equality because Ψ_3 is faithful. $\square$

5. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Fix $j \geq 3$. Because all three letters occur in u , the word u^j contains the decreasing subsequence 321: choose 3 from the first copy, 2 from the second, and 1 from the third.

Suppose first that $w \in C(u^j)$. The row-bound proposition cited in Section 2 shows that the first three rows of $P(w)$ contain no letter larger than 3. Lemma 2.1 and Corollary 4.2 give

$$w_{\leq 3} \in C_{\text{Plac}_3}(u^j) \iff w_{\leq 3} \in C_{\text{Plac}_3}(u^3).$$

Applying Lemma 2.1 again gives $w \in C(u^3)$. Thus $C(u^j) \subseteq C(u^3)$.

Conversely, start with $w \in C(u^3)$. The same 321 row bound applies to u^3 , so the same sequence of tail isolation, Corollary 4.2, and tail isolation gives $w \in C(u^j)$. Hence $C(u^3) \subseteq C(u^j)$. $\square$

6. THE INDEPENDENT SIX-COORDINATE PRODUCT

Although the tropical proof does not require coordinates, we record and prove the exact rank-three product formula used in the independent structural route and in one finite enumerator. This also fixes all boundary cases of the row-word convention.

For nonnegative integers a, b, c, d, e, f , put

$$T(a, b, c, d, e, f) = \frac{1^a 2^b 3^c}{2^d 3^e f}.$$

Proposition 6.1. *The semistandard tableaux with entries in $[3]$ are in bijection with the nonnegative sextuples satisfying*

$$f \leq d \leq a, \quad d + e \leq a + b.$$

Proof. A strictly increasing column over $[3]$ has at most three entries. Weak row increase and strict column increase force the first row to have the displayed form. No 1 can occur in row 2, since it would require a smaller positive entry above it, so row 2 is $2^d 3^e$. A row-3 entry has two strictly smaller positive entries above it, so it must be 3 and the entry immediately above it must be 2. Thus row 3 is 3^f , and all six multiplicities are unique. A row-2 entry 2 must lie below a 1, giving $d \leq a$; every row-2 entry must lie below a 1 or 2, giving $d + e \leq a + b$; and a row-3 entry 3 must lie below a row-2 entry 2, giving $f \leq d$.

Conversely, these inequalities imply $f \leq d \leq d + e \leq a + b \leq a + b + c$. They place the row-2 2s below 1s, the row-2 3s below 1s or 2s, and all row-3 3s below row-2 2s. Hence the display is an SSYT. $\square$

Lemma 6.2 (row word). *For every feasible sextuple,*

$$P(3^f 2^d 3^e 1^a 2^b 3^c) = T(a, b, c, d, e, f).$$

Proof. This is a direct insertion from the empty tableau. The first three batches $3^f, 2^d, 3^e$ give the rows $2^d 3^e / 3^f$, because $f \leq d$. Put $g = \min(a - d, e)$, which is defined since $d \leq a$. Inserting 1^a bumps the word $2^d 3^g$ from row 1. The batch 2^d bumps all f row-2 3s, and 3^g then appends there, so the resulting rows are

$$1^a 3^{e-g} / 2^d 3^g / 3^f.$$

When 2^b is inserted, it bumps all $e - g$ remaining row-1 3s. Indeed, this is immediate if $g = e$; otherwise $g = a - d < e$, and $d + e \leq a + b$ gives $e - g = e - a + d \leq b$. Those 3s append to row 2, producing $1^a 2^b / 2^d 3^e / 3^f$. The final batch 3^c appends to row 1. This proves the claimed representative without using the product formula. $\square$

Let $S = T(a, b, c, d, e, f)$ and $T = T(A, B, C, D, E, F)$ be feasible. Define

$$\begin{aligned} r &= \min(D, c + F), & q &= \min(A, b + D), \\ t &= \min(q, e + r), & v_0 &= \min(A - q, c + F - r + E), \\ (21) \quad \ell &= \min(B, c + F - r + E - v_0). \end{aligned}$$

Proposition 6.3 (six-coordinate product). *The plactic product is*

$$(22) \quad S \cdot T = T(a + A, b + D - q + B, c + F - r + E - v_0 - \ell + C, \\ d + q, e + r + v_0 - t + \ell, f + t).$$

Proof. By Lemma 6.2, insert the row word $3^F 2^D 3^E 1^A 2^B 3^C$ into S . The following batch induction uses only that row insertion bumps the leftmost strictly larger entry.

First, 3^F appends to row 1. On inserting 2^D , exactly $r = \min(D, c + F)$ row-1 3s are bumped; the bumped word 3^r appends to row 2. After the following 3^E appends, put $c_1 = c + F - r + E$. The complete state is

$$(23) \quad \begin{array}{c} 1^a 2^{b+D} 3^{c_1} \\ 2^d 3^{e+r} \\ 3^f \end{array}.$$

Now insert 1^A . The first $q = \min(A, b + D)$ effective insertions bump the row-1 2s. After those are exhausted, exactly $v_0 = \min(A - q, c_1)$ further insertions bump row-1 3s; any remaining 1s append. Thus row 1 becomes

$$1^{a+A} 2^{b+D-q} 3^{c_1-v_0},$$

and the bumped word in its actual order is $2^q 3^{v_0}$. When 2^q is inserted into row 2 of (23), exactly $t = \min(q, e + r)$ row-2 3s are bumped to row 3, while

any remaining incoming 2s append. The following 3^{v_0} appends to row 2. Hence the state after the entire 1^A batch is

$$(24) \quad \begin{matrix} 1^{a+A} 2^{b+D-q} 3^{c_1-v_0} \\ 2^{d+q} 3^{e+r+v_0-t} \\ 3^{f+t}. \end{matrix}$$

This line accounts separately for all six current coordinates.

Finally, inserting 2^B bumps exactly $\ell = \min(B, c_1 - v_0)$ row-1 3s, after which the remaining 2s append; the bumped 3^ℓ appends to row 2. Inserting 3^C then appends to row 1. The resulting rows are

$$\begin{matrix} 1^{a+A} 2^{b+D-q+B} 3^{c_1-v_0-\ell+C} \\ 2^{d+q} 3^{e+r+v_0-t+\ell} \\ 3^{f+t}, \end{matrix}$$

which is (22) after substituting c_1 . At each individual insertion the indicated multiplicity decreases by one until exhausted and then the incoming letter appends, so the argument is an induction on the size of each batch, not a simultaneous heuristic. Since every state is obtained by literal row insertion, the final sextuple is feasible.

For an explicit empty-factor endpoint check, set the coordinates of S to zero. Feasibility of T gives

$$r = F, \quad q = D, \quad t = F, \quad v_0 = \min(A - D, E).$$

Also $\ell = E - v_0$: if $E \leq A - D$ both sides vanish, while otherwise $v_0 = A - D$ and $D + E \leq A + B$ gives $E - v_0 = E - A + D \leq B$. Thus (22) reduces exactly to (A, B, C, D, E, F) , including all zero and equality cases. $\square$

7. EXACT FINITE VERIFICATION

The proof above is human and does not depend on computation. As an adversarial check, we implemented ordinary row insertion for discovery and a separate verifier that enumerates semistandard tableaux by shape and filling and evaluates products using column insertion of the reversed word. The verifier imports no discovery code, reconstructs the complete membership-bit stream from sealed parameters, checks SHA-256 digests, and rejects malformed or tampered inputs.

Enumeration by one representative per tableau is exhaustive in every stated finite box because Knuth equivalence is a congruence and content is constant on a plactic class. The first row of the table reproduces the $m = 3$ component of the finite experiment reported in [2]. The other rows are separate boxes and must not be combined into an uncomputed rectangle.

$ u $	witnesses	powers compared	class pairs	comparisons
$3 \dots 8$	$w \in [3]^{\leq 10}$	$3 \dots 14$	872,298	10,467,576
$3 \dots 8$	$w \in [5]^{\leq 7}$	$3 \dots 20$	3,724,896	67,048,128
$3 \dots 10$	$w \in [3]^{\leq 10}$	$3 \dots 20$	2,447,130	44,048,340
$3 \dots 8$	$w \in [3]^{\leq 16}$	$3 \dots 20$	6,869,709	123,654,762

Every comparison count is exact and every run found zero adjacent membership changes. The source bundle includes all four bit streams, result records, per-run manifests, independent verification logs, and rejection tests. Wall times are recorded only as diagnostics and are not part of any claim.

COMPUTATIONAL AND PROOF-ASSISTANT DISCLOSURE

The theorem is proved by the written combinatorial and tropical arguments in Sections 2–5. Exact finite enumeration and SMT experiments were used only for discovery and adversarial testing; no solver response is used as a proof step. No Lean, Coq, Isabelle, or other interactive proof assistant was used, and no formalization claim is made. The finite verifier uses only integer and combinatorial operations and is fail-closed on malformed serialized input.

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